

TrES-1b

R-1395

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_130507 · created 2026-08-01T03:25:50+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456419.89375 +/- 0.00084 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1408 +/- 0.0064
Transit depth (Rp/Rs)^2	1.98 +/- 0.18 [%]
Orbital Inclination (inc)	89.11 +/- 0.61
Airmass coefficient 1 (a1)	1.1649 +/- 0.0099
Airmass coefficient 2 (a2)	-0.1358 +/- 0.0064
Scatter in the residuals of the lightcurve fit is	0.86 %
Best Comparison Star	None
Optimal Aperture	4.3
Optimal Annulus	8.93
Transit Duration (day)	0.1031 +/- 0.0017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1408 ± 0.0064 vs 0.1358 (deviation 0.73σ)
Duration fit vs published	0.1031 d vs 0.1042 d (deviation 0.59σ)
Mid-time O–C	2.6 min
Depth SNR	20.6
Residual scatter	0.86 %

Light curve

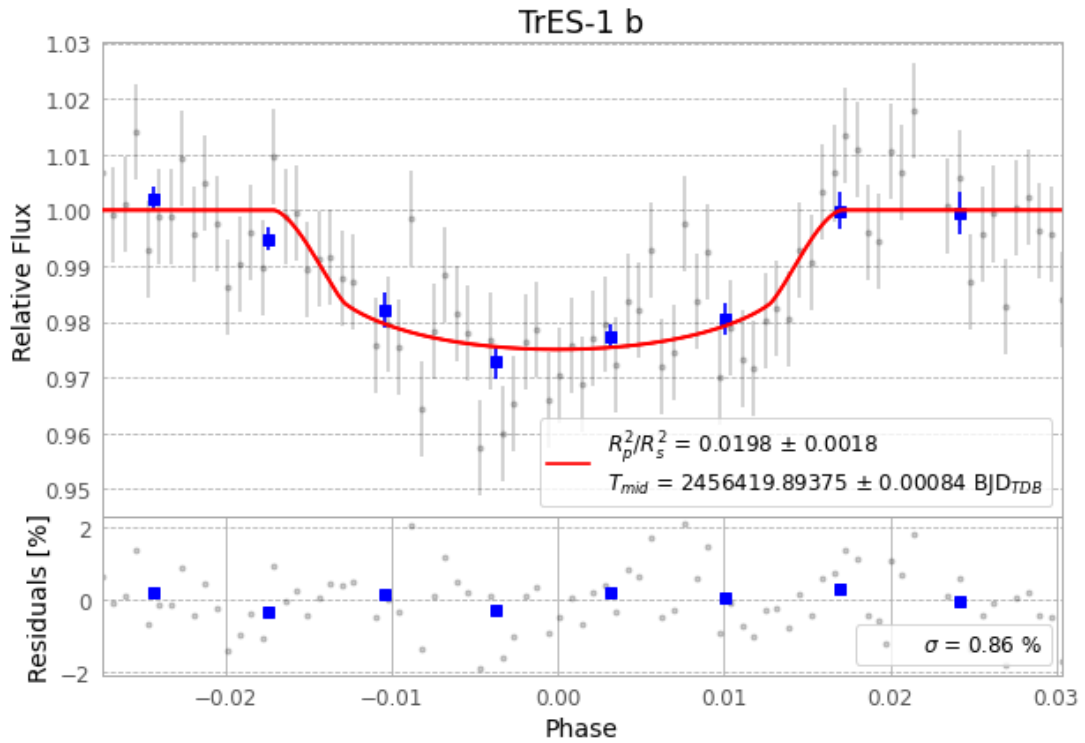


Figure 1 — FinalLightCurve_TrES-1 b_2013-05-07.png (SHA-256 abb034e8efd3e730...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [338, 202], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b0f3ce7c8f5426b7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), TRES-1130507072412.FITS → TRES-1130507113612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-130507.log (eedec516cb70...), FinalLightCurve_TrES-1 b_2013-05-07.png (abb034e8efd3...), FinalLightCurve_TrES-1 b_2013-05-07.csv (33195312f5ef...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 03:25Z.